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| STANDARD OPERATING PROCEDURE                | MultiLink.Trial.SOP.17 v.1   |
| Title: Patient linkage to community support |                              |
| Effective date: 7 AUG. 2025                 | Next Review Due: 7 AUG. 2026 |

|          | Name and position                             |
|----------|-----------------------------------------------|
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#### REVISION HISTORY

| Ver. No. | Revised by (Name & Position) | Effective Date | Details of changes                                                 |
|----------|------------------------------|----------------|--------------------------------------------------------------------|
| 0.2      | Jennifer Ljungqvist          | 17/07/2025     | Minor edits to 4.2.3 text and appendix 1 (as discussed with Felix) |
| 0.3      | Jennifer Ljungqvist          | 29/07/2025     | Minor edits to align SOP with eCRF questions and updated process   |
| 1        | Jennifer Ljungqvist          | 05/08/2025     | Edits in response to EW review                                     |

## **1 BACKGROUND**

This Standard Operating Procedure (SOP) has been developed to streamline the process of delivering study participant/patient linkage to community support for ongoing management of multimorbidity for participants enrolled in the six intervention sites in the Multilink external trial.

## **2 PURPOSE**

- 2.1 To outline the processes of linking participants to community support for ongoing management of multimorbidity in home and community settings after discharge from hospitals.
- 2.2 To describe the resources needed for the efficient delivery of linking participants to community support for ongoing management of multimorbidity.
- 2.3 To detail approaches to evaluating adherence to SOP and effectiveness of linking patients to community support for ongoing management of multimorbidity

## **3 RESPONSIBILITY**

Research nurses in intervention hospitals have the primary responsibility of delivering participant linkage to community support and adhering to its protocol. However, each member of the study team must ensure adherence to protocols that outline the intervention to link participants to community support for ongoing management of multimorbidity. The study coordinators, site PIs and research nurses will all assist with the mapping of community organisations/structures/networks and services.

## **4 PROCEDURE**

### **4.1 Materials**

- Mapped list of community structures to link participants to for ongoing support after hospital discharge.
- List of contact information of individuals in the district hospitals, primary healthcare facilities (PHC), and other community structures responsible for community support, logged in a Register of Community Structures for Participant Linkage
- Materials for participant education for self-management (booklet, flipbook, poster)
- An evaluation form to assess the education of the existing community organisations.
- The 'Participant Level Interventions' forms within the electronic Case Report Form (eCRF) in the trial database on REDCap, to assess the delivery of the community linkage intervention and capture participants' use of community structures.

### **4.2 Methods**

The flowchart Appendix 1 (9.1) and topic guide in Appendix 2 (9.2) of this SOP describe the process of linking participants to community support for ongoing management of multimorbidity in the intervention arm. While the community linkage intervention will only be delivered in the districts linked to the six intervention sites, community support structures will be mapped for all districts where the 12 study hospitals are located (Appendix 3, section 9.3). Below is a step-by-step procedure for delivering participant linkage to community support.

#### **4.2.1. Mapping of community structures**

To create the materials needed for the community linkage intervention in the hospital, research staff in each district where the 11 study sites are located need to first map out all known community structures/networks/organisations that operate and provide diabetes MultiLink.Trial.SOP.17 v.1

mellitus (DM), hypertension (HPT) and/or HIV/Aids community services in those catchment areas. This mapping could be done through the following strategies:

1. Multilink has partnered with the **NCD Alliance** for this project. We will engage its national offices to identify its community-based membership in the catchment areas of the trial study hospitals.
  2. The study coordinators and research nurses with possible support from the site PIs (Principal Investigators) will work with the **district hospital** to establish the existence of an office/individual within the district hospital that collaborates directly with community structures and map out these structures.
  3. The study coordinators and research nurses with possible support from staff within the study hospitals will map **PHC facilities** in the catchment areas of the district hospitals that are included in the external trial and randomised to the intervention arm. They will aim to identify offices/individuals responsible for outreach activities in these PHC facilities.
  4. The study coordinators and research nurses with possible support from site PIs will further engage with Community Advisory Board (CAB) members, community leaders, Ministry of Health contacts and any other networks as needed to obtain an exhaustive list of DM, HPT and HIV/Aids community support services provided in the study districts.
- Community structures identified through the four approaches described above will be mapped out to establish specific areas where they operate, and their contact details (locations, phone numbers, group leaders) will be logged in a Register of Community Structures for Participant Linkage.
  - Once complete, the Register will serve as the primary sources for producing district-specific contact lists for community organisations that can be distributed to participants enrolled at the six intervention sites.

#### **4.2.2. Training of community structures in multimorbidity and participant linkage**

- Representatives of the community structures mapped in the catchment areas of the *six intervention hospitals* will be invited to a daylong orientation session about the project (maximum 30 participants) covering the **aim of the study**, the aspects of **linking participants to community support** and **safeguarding** of the research participants. The representatives of the community structures (including those from community organizations and PHC facilities) will also be trained in the basics of **multimorbidity** and its management. The content in the patient education materials can be referenced during this training.
- The training will be facilitated by the Multilink clinical leads and supported by the study coordinators who are in charge of the study sites in each country. The study coordinator is responsible for running the trial at each site, including making sure that all supplies and equipment is in place, training has been conducted and that implementation of the trial goes according to plan. The clinical leads are doctors who provide clinical support for the trial in each country.
- This orientation will happen at district level in local national languages (Chichewa for Malawi and Kiswahili for Tanzania). Materials for delivering the orientation session will include Multilink participant education materials for self-management (i.e. poster, flipbook, booklet).

#### **4.2.3. Participant linkage to ongoing community support for multimorbidity management**

### *Initial Visit (Inpatient Admissions)*

- Prior to discharge, research nurses will engage the participant and their caregivers to understand where they live/reside. The nurses will identify (from the Register of Community Structures for Participant Linkage) any community structures in NCD care that are near (within five-mile radius or 30 minutes of travel) the participant's residence.
- Research nurses will provide the participant with a printed copy of the contact details for the community structures in their district. In conversation with the participant and/or their caregiver, the research nurses will then point out the contact details of organizations that are closest to the participant's residence and the person responsible for outreach activities for their usual PHC facility.
  - This delivery of community linkage information can take place in conjunction with the patient/participant education intervention outlined in '10. MultiLink SOP\_patient\_education\_v1.0'
- Research nurses will also make note of any community structures within the district that the participant has used previously, which are not already in the Register, and record this information through the eCRF on REDCap, using the 'Participant Level Interventions (Initial Visit)' form.
- After linking the participants with community organisations and PHC facilities it is up to the participants to reach out to those contacts and for the participants and community representatives to interact as they see fit.

### *Follow-up Visits (Day 30, 60, 90)*

- When the study participants enrolled return to the intervention hospital for their follow-up visits around days 30, 60 and 90 of their enrolment in the trial, the research nurse will discuss their use of community support services/structures since their last visit.
- These conversations can take place in conjunction with the patient/participant re-education sessions on self-management of multimorbidity. During these sessions, the research nurse will collect data from the participant and/or their caregiver through the 'Participant Level Interventions (Follow-up Visits)' form in the eCRF in REDCap.
- After meeting with the participant and/or their caregiver, the research nurse will provide feedback on the follow-up education and community linkage session held in the second part of the 'Participant Level Interventions (Follow-up Visits)' form in REDCap.

## **5 HEALTH AND SAFETY**

The health and safety section in the SOP focuses on ensuring the physical and psychological well-being of participants during the process. This includes:

1. Safety Protocols: Adherence to safety guidelines to protect participants and staff, especially in clinical settings. This includes the use of Personal Protective Equipment (PPE) as needed.
  - Community organisations that are trained on how to provide ongoing support will also be trained on safeguarding measure when interacting with vulnerable participants
  - Research nurses will only link participants to community organisations (share this information) if they want to be linked.
  - The identities of study participants will not be shared with community organisations. It is up to participants if they want to reach out to community organisations identified or not.

2. Confidentiality: Strict maintenance of confidentiality for all participant information gathered during the process.
3. Discomfort mitigation: Procedures to minimize any stress or discomfort for participants including immediate response to any health concerns.
4. Safe Environment: Conducting in a private and comfortable setting.

## **6 ASSOCIATED PROCEDURES**

- Education sessions on the self-management of multimorbidity
- Randomisation of study sites into standard (control) and intervention arms

## **7 REFERENCES**

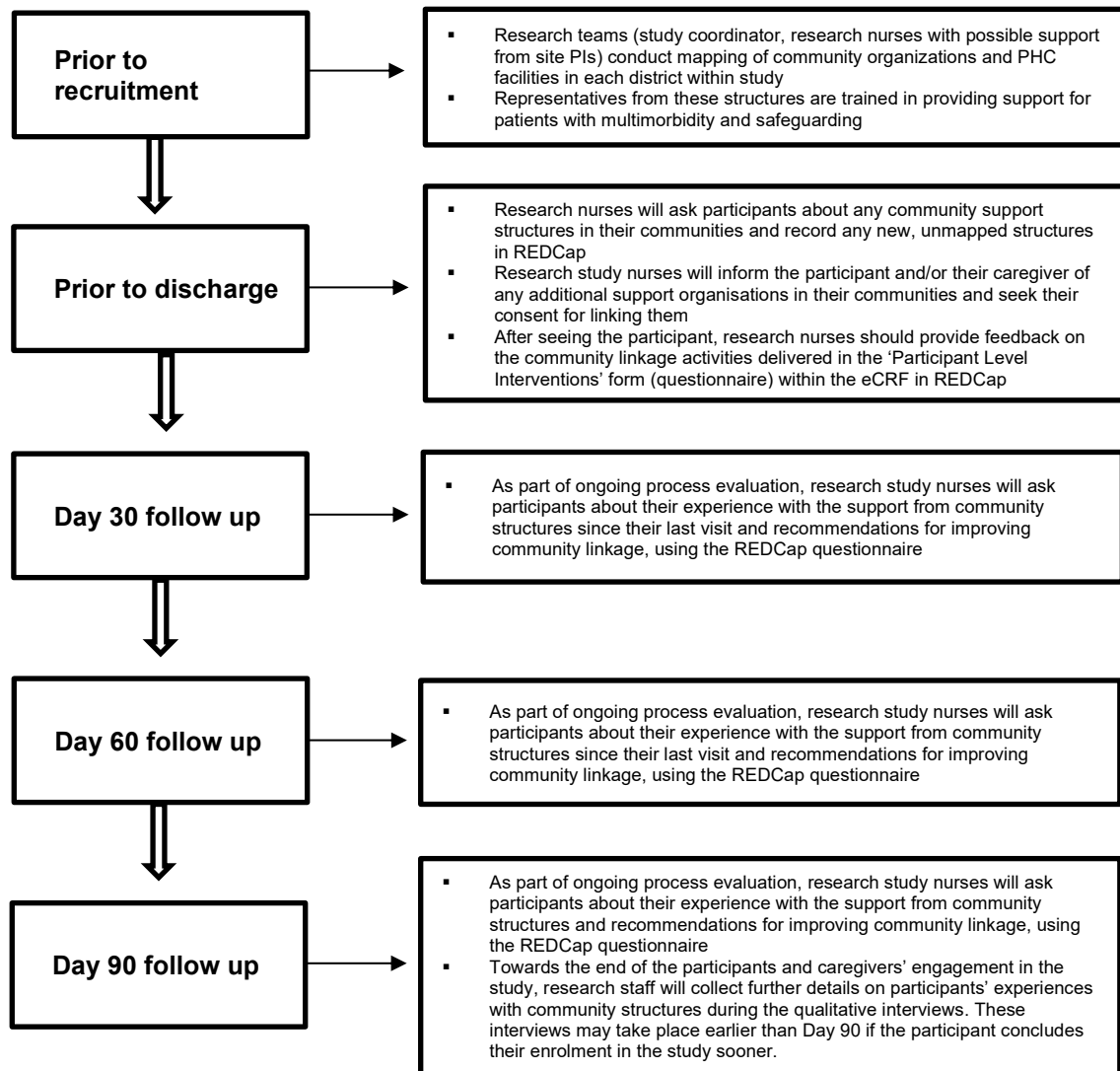
- Multilink\_Trial\_Protocol\_v5.0\_02May2025
- 10. MultiLink SOP\_patient\_education\_v1.0
- MLW.SOP.MLINK.011\_randomisation\_v2.0\_clean\_signed

## **8 LIST OF ABBREVIATION**

|      |                               |
|------|-------------------------------|
| CAB  | Community Advisory Board      |
| CI   | Co-Investigator               |
| DM   | Diabetes Mellitus             |
| eCRF | electronic Case Report Form   |
| HCW  | Healthcare Workers            |
| HPT  | Hypertension                  |
| NCD  | Non-communicable disease      |
| PHC  | Primary Health Care           |
| PI   | Principal Investigator        |
| PPE  | Personal Protective Equipment |
| SOP  | Standard Operating Procedure  |

## 9 APPENDICES

### 9.1 Appendix 1: Intervention arm flow chart of delivering participant linkage to community support for ongoing multimorbidity management



## 9.2 Appendix 2: Flow of contents for participant linkage to community support for ongoing multimorbidity management

| Topic to Discuss                                        | Study Nurse and Participant to Talk About                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Community Structures Used</b>                        | <ol style="list-style-type: none"> <li>1. Address/Resident of Participant</li> <li>2. Participant's awareness of any community support services/structures nearby</li> <li>3. Used any community structures previously (e.g. primary healthcare facilities, other community services, visits by community outreach representatives)</li> </ol>                                                                      |
| <b>Community Linkage</b>                                | <ol style="list-style-type: none"> <li>1. Offer participant further information on community structures that Multilink has identified/mapped/trained</li> <li>2. Highlight structures close to participant's home address</li> </ol>                                                                                                                                                                                |
| <b>Community Structures Feedback (from Participant)</b> | <p>[Only at follow-up visits D30, D60, D90]</p> <ol style="list-style-type: none"> <li>1. Used any community structures previously recommended</li> <li>2. What community structures used</li> <li>3. Challenges in accessing community structures</li> <li>4. Type of support received from community structures and views on this support</li> <li>5. How community support services could be improved</li> </ol> |

## 9.3 Appendix 3: Study sites and districts

| Hospitals (Sites)                       | District                                 | Study Team                     | PI (main)              |
|-----------------------------------------|------------------------------------------|--------------------------------|------------------------|
| Malawi                                  |                                          |                                |                        |
| 1. Chiradzulu District Hospital         | Chiradzulu district                      | MLW,<br>Blantyre               | Dr Steve<br>Kateta     |
| 2. Mwanza District Hospital             | Mwanza district                          |                                |                        |
| 3. Nsanje District Hospital             | Nsanje district                          |                                | Dr Mulinda<br>Nyirenda |
| 4. Ntcheu District Hospital             | Ntcheu district                          |                                |                        |
| Tanzania                                |                                          |                                |                        |
| 5. Amana Hospital                       | Ilala district                           | MUHAS, Dar<br>es Salaam        | Prof Henry<br>Sawe     |
| 6. Kigamboni District Hospital          | Somangila ward,<br>Kigamboni district    |                                |                        |
| 7. Mwananyamala District<br>Hospital    | Mwananyamala,<br>Kinondoni district      |                                |                        |
| 8. Temeke Regional Referral<br>Hospital | Temeke district                          |                                |                        |
| 9. Hai District Hospital                | Hai district                             | KCMC,<br>Kilimanjaro<br>region | Dr. Sarah<br>Urasa     |
| 10. Machame District Hospital           | Hai district                             |                                |                        |
| 11. Kibosho District Hospital           | Moshi Rural District                     |                                |                        |
| 12. St. Joseph District Hospital        | Moshi Municipality<br>(South West Moshi) |                                |                        |

## 10 TRAINING RECORD FOR THIS SOP

The following people have been trained in this SOP and have demonstrated adequate knowledge of its contents:

[illegible]